

Sarthak kumar seth

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EDUCATION

National Institute of Technology Delhi — New Delhi, Delhi

Bachelor of Technology in Electrical Engineering

Graduating 2027

The Srijan School — New Delhi, Delhi

Senior Secondary

Completed June 2023

WORK EXPERIENCE

Summer Intern – DEAL DRDO, Dehradun

June 2025 - July 2025

- Developed a **Machine Learning-based State of Charge (SoC) estimation** system for Li-ion batteries, integrating data-driven models with **Extended Kalman Filter (EKF)** baselines.
- Built, trained, and optimized ML models using **scikit-learn** and deep learning frameworks, including **Linear Regression, Decision Trees, Random Forest, Gradient Boosting, Neural Networks, and LSTM**.
- Evaluated model performance using standard regression metrics such as **RMSE, MAE, and R^2** , and visualized estimation errors and trends using **Matplotlib** and **Seaborn**.

PROJECTS

Smart Expense Tracker | Smart Expense Tracker | 

- Engineered a full-stack finance tracking platform with secure Google OAuth 2.0 authentication, enabling multi-user income and expense management with structured category classification.
- Developed a rule-base budget monitoring engine with automated SMTP email alerts on threshold breach, reducing manual tracking effort by 60 and improving proactive budget control.
- Built scalable RESTful APIs and an interactive analytics dashboard generating monthly financial reports, category-wise expenditure insights, and optimized PostgreSQL queries using indexed schema design.
- Technologies:** Python, Django, Django REST Framework, PostgreSQL, Google OAuth, SMTP, Git

Battery State of Charge Estimation System | 

- Processed 50k+ time-series samples to estimate battery charge under dynamic operating conditions.
- Trained and compared 5+ models including ensemble methods and LSTM, achieving up to 18% RMSE reduction.
- Validated performance using RMSE, MAE, and R^2 across multiple charge-discharge cycles.
- Technologies:** Machine Learning, Python, scikit-learn, LSTM, jupyter Notebook

Ice Cream E-Commerce Management System | 

- Built and deployed a scalable multi-role e-commerce platform (Django, Render) handling vendor/customer workflows and managing 100+ products with modular backend architecture.
- Designed normalized relational models and RESTful APIs for Users, Vendors, Products, and Orders, leveraging atomic transactions to ensure data consistency and handle concurrent checkout scenarios.
- Implemented end-to-end order pipeline (cart → checkout → order) with real-time inventory updates, reducing overselling risk by 90 and improving system reliability under concurrent usage.
- Technologies:** Python, Django, REST API, OAuth

TECHNICAL SKILLS

Programming Languages: Python, C++, JavaScript, SQL, MATLAB

Backend & Web Frameworks: Django, Flask

Frontend Technologies: HTML5, CSS3, Tailwind CSS, JavaScript, Next.js

Machine Learning: scikit-learn, TensorFlow, LSTM, XGBoost, Decision Trees, Neural Networks

Computer Science Fundamentals: DSA, Object-Oriented Programming (OOP), OS

Developer Tools: Git, GitHub, VS Code, Jupyter Notebook

EXTRA-CURRICULAR

Training & Placement Cell, NIT Delhi — Database Head

Dec 2025 – Present

Unnat Bharat Abhiyan (UBA) Cell, NIT Delhi — General Secretary

Aug 2024 – Present